A	B	C	D	E	F	G	H	I	J
AdCampaignID	**AdCampaign Name**	**StartDate**	**Duration**	**Campaign MgrID**	**Campaign MgrName**	**ModusID**	**Media**	**Range**	**Budget Pctg**
111	SummerFun20	6.6.2020	12 days	CM100	Roberta	1	TV	Local	50%
111	SummerFun20	6.6.2020	12 days	CM100	Roberta	2	TV	National	50%
222	SummerZing20	6.8.2020	30 days	CM101	Sue	1	TV	Local	60%
222	SummerZing20	6.8.2020	30 days	CM101	Sue	3	Radio	Local	30%
222	SummerZing20	6.8.2020	30 days	CM101	Sue	5	Print	Local	10%
333	FallBall20	6.9.2020	12 days	CM102	John	3	Radio	Local	80%
333	FallBall20	6.9.2020	12 days	CM102	John	4	Radio	National	20%
444	AutmnStyle20	6.9.2020	5 days	CM103	Nancy	6	Print	National	100%
555	AutmnColors20	6.9.2020	3 days	CM100	Roberta	3	Radio	Local	100%

Initially recognized sets of functional dependencies in the relation AD CAMPAIGN MIX

(Set 1) AdCampaignID → AdCampaignName, StartDate, Duration, CampaignMgrID, CampaignMgrName

(Set 2) AdCampaignName → AdCampaignID, StartDate, Duration, CampaignMgrID, CampaignMgrName

(Set 3) CampaignMgrID → CampaignMgrName

(Set 4) ModusID → Media, Range

(Set 5) AdCampaignID, ModusID → AdCampaignName, StartDate, Duration, CampaignMgrID, CampaignMgrName, Media, Range, BudgetPctg

(Set 6) AdCampaignName, ModusiD → AdCampaignID, StartDate, Duration, CampaignMgrID, CampaignMgrName, Media, Range, BudgetPctg

Set 1: → A → BCDEF

Set 2: B → ACDEF

Set 3: E → F

Set 4: G → HI

Set 5: AG → BCDEFHIJ

Set 6: BG → ACDEFHIJ

NOW

✓ Set1: $A \rightarrow BCDEF$

Set3: $E \rightarrow F$

Set4: $G \rightarrow HI$

✓ Set5: $AG \rightarrow BCDEFHIJ$

$AG \rightarrow HIJ$

$AG \rightarrow J$

$A \rightarrow B$

X $AG \rightarrow B$

$G \rightarrow H$

X $AG \rightarrow H$

Set1: $A \rightarrow BCDEF$

Set3: $E \rightarrow F$

Set4: $G \rightarrow HI$

Set5: $AG \rightarrow J$

Set1 $A \rightarrow BCDEF$ X

X Set2 $B \rightarrow ACDEF$

$A \rightarrow Bx$

$B \rightarrow Ax$

NOW

Set1: $A \rightarrow BCDEF$

Set3: $E \rightarrow F$

Set4: $G \rightarrow HI$

✓ Set5: $AG \rightarrow BCDEFHIJ$ Y

X ✓ Set6: $BG \rightarrow ACDEFHIJ$

$AG \rightarrow BY$

$BG \rightarrow AY$ ✓

AB

{A} {B}

PA = {A, B, D}

CK = {AB, DB}

{AC}$^+$ = {ACDEFB}

So AC is a SK.

A^+ = A Not SK

C^+ = {CDEFAB}

C is a SK.

C is also CD.

PA = {A, B, D, C}

CK = {AB, DB, C}

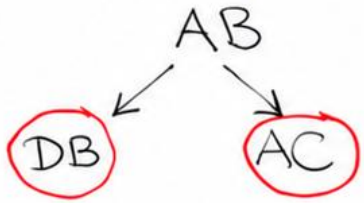
Finding Candidate Keys and Super Keys of a Relation using FDs

So AB is a SK.

So AB is also CK.

PA = {A, B}

CK = {AB}



{DB}$^+$ = {DBACEF}

So DB is a SK.

D^+ = DA. Not SK

B^+ = B Not SK

So DB is a CK.

Consider a relation

R(A,B, C, D, E, F) With the following FDs:

AB → C
C→ D
C→ E
E→ F
D→ A
C→B

Find all possible candidate keys?

Solution

{ABCDEF}$^+$ = {ABCDEF}

{ABDEF}$^+$ = {ABCDEF}

{ABEF}$^+$ = {ABCDEF}

{ABF}$^+$ = {ABCDEF}

{AB}$^+$ = {ABCDEF}